

Gamma-Function Representations of Selected Algebraic Numbers

Thomas Scheuerle

<https://www.linkedin.com/in/scheuerle/>

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Abstract

We collect explicit representations of algebraic numbers as products and quotients of Gamma values at rational arguments. For each entry, a minimal polynomial and a corresponding real root are given. The identities were discovered computationally and are presented as a reference table. No claim of completeness is made.

1 Introduction

The Gamma function admits numerous special values and functional identities. Ratios of Gamma values at rational arguments frequently simplify to algebraic expressions through the reflection, multiplication and related formulas. This note records a collection of experimentally discovered examples in which real algebraic numbers are represented by finite products or quotients of Gamma values evaluated at rational arguments. For each entry, the minimal polynomial and an associated real root are listed. The collection is intended primarily as a reference resource and as a source of examples for further investigations into special values of the Gamma function.

Relation to Trigonometric Values

Many entries can be rewritten using Euler's reflection formula

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}.$$

Consequently, numerous Gamma-function quotients appearing in this catalogue are equivalent to ratios of trigonometric values at rational multiples of π . Such quantities belong to cyclotomic number fields and are therefore algebraic. The identities recorded in this catalogue may thus be viewed as explicit Gamma-function realizations of algebraic numbers arising from cyclotomic constructions.

Conventions

For each entry, a polynomial $P(x) \in \mathbb{Z}[x]$ and a Gamma-function expression R are displayed in the form

$$P(x) = 0 \implies x = R,$$

meaning that the quantity R is a real root of P . Only one root and one corresponding representation are recorded for each entry. Additional roots or alternative Gamma-function representations may exist and are not considered here. The purpose of this catalogue is to document explicit evaluations connecting algebraic numbers with products and quotients of Gamma values at rational arguments. The collection is intended as a reference resource and no claim of completeness is made.

2 Quadratic Equations

$$x^2 - 2 = 0 \Rightarrow x = \frac{\Gamma(1/4)\Gamma(3/4)}{\Gamma(1/2)^2} \quad (1)$$

$$x^2 - 2 = 0 \Rightarrow x = \frac{\Gamma(1/6)\Gamma(5/6)}{\Gamma(1/4)\Gamma(3/4)} \quad (2)$$

$$2x^2 - 1 = 0 \Rightarrow x = \frac{\Gamma(1/2)^2}{\Gamma(1/4)\Gamma(3/4)} \quad (3)$$

$$2x^2 - 1 = 0 \Rightarrow x = \frac{\Gamma(1/4)\Gamma(3/4)}{\Gamma(1/6)\Gamma(5/6)} \quad (4)$$

$$x^2 - x - 1 = 0 \Rightarrow x = \frac{\Gamma(1/5)\Gamma(4/5)}{\Gamma(2/5)\Gamma(3/5)} \quad (5)$$

$$x^2 + x - 1 = 0 \Rightarrow x = \frac{\Gamma(2/5)\Gamma(3/5)}{\Gamma(1/5)\Gamma(4/5)} \quad (6)$$

$$x^2 - 3 = 0 \Rightarrow x = \frac{\Gamma(1/6)\Gamma(5/6)}{\Gamma(1/3)\Gamma(2/3)} \quad (7)$$

$$3x^2 - 1 = 0 \Rightarrow x = \frac{\Gamma(1/3)\Gamma(2/3)}{\Gamma(1/6)\Gamma(5/6)} \quad (8)$$

$$3x^2 - 4 = 0 \Rightarrow x = \frac{\Gamma(1/3)\Gamma(2/3)}{\Gamma(1/2)^2} \quad (9)$$

$$4x^2 - 3 = 0 \Rightarrow x = \frac{\Gamma(1/2)^2}{\Gamma(1/3)\Gamma(2/3)} \quad (10)$$

$$x^2 - 2x - 1 = 0 \Rightarrow x = \frac{\Gamma(1/8)\Gamma(7/8)}{\Gamma(3/8)\Gamma(5/8)} \quad (11)$$

$$x^2 + 2x - 1 = 0 \Rightarrow x = \frac{\Gamma(3/8)\Gamma(5/8)}{\Gamma(1/8)\Gamma(7/8)} \quad (12)$$

$$x^2 - 2x - 4 = 0 \Rightarrow x = \frac{\Gamma(1/10)\Gamma(9/10)}{\Gamma(1/2)^2} \quad (13)$$

$$x^2 - 3x + 1 = 0 \Rightarrow x = \frac{\Gamma(3/10)\Gamma(7/10)}{\Gamma(1/10)\Gamma(9/10)} \quad (14)$$

$$x^2 + 2x - 4 = 0 \Rightarrow x = \frac{\Gamma(3/10)\Gamma(7/10)}{\Gamma(1/2)^2} \quad (15)$$

$$4x^2 + 2x - 1 = 0 \Rightarrow x = \frac{\Gamma(1/2)^2}{\Gamma(1/10)\Gamma(9/10)} \quad (16)$$

$$4x^2 - 2x - 1 = 0 \Rightarrow x = \frac{\Gamma(1/2)^2}{\Gamma(3/10)\Gamma(7/10)} \quad (17)$$

$$x^2 - 5 = 0 \Rightarrow x = \frac{\Gamma(1/10)^2\Gamma(3/10)\Gamma(1/2)}{\Gamma(1/5)^3\Gamma(2/5)} \quad (18)$$

$$5x^2 - 1 = 0 \Rightarrow x = \frac{\Gamma(1/5)^3\Gamma(2/5)}{\Gamma(1/10)^2\Gamma(3/10)\Gamma(1/2)} \quad (19)$$

$$x^2 - 2x - 2 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/12)\Gamma(11/12)}{\Gamma(1/4)\Gamma(3/4)} \quad (20)$$

$$x^2 - 2x - 2 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/12)\Gamma(2/3)}{\Gamma(1/3)\Gamma(5/12)} \quad (21)$$

$$x^2 - 4x + 1 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/12)\Gamma(11/12)}{\Gamma(5/12)\Gamma(7/12)} \quad (22)$$

$$x^2 - 4x + 1 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(5/12)\Gamma(7/12)}{\Gamma(1/12)\Gamma(11/12)} \quad (23)$$

$$2x^2 + 2x - 1 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/4)\Gamma(3/4)}{\Gamma(1/12)\Gamma(11/12)} \quad (24)$$

$$2x^2 + 2x - 1 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/3)\Gamma(5/12)}{\Gamma(1/12)\Gamma(2/3)} \quad (25)$$

$$2x^2 - 3 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/4)\Gamma(3/4)}{\Gamma(1/3)\Gamma(2/3)} \quad (26)$$

$$2x^2 - 2x - 1 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/4)\Gamma(3/4)}{\Gamma(5/12)\Gamma(7/12)} \quad (27)$$

$$3x^2 - 2 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/3)\Gamma(2/3)}{\Gamma(1/4)\Gamma(3/4)} \quad (28)$$

$$x^2 + 2x - 2 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(5/12)\Gamma(7/12)}{\Gamma(1/4)\Gamma(3/4)} \quad (29)$$

$$x^2 - 6 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/12)\Gamma(7/12)}{\Gamma(1/3)^2} \quad (30)$$

$$6x^2 - 1 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/3)^2}{\Gamma(1/12)\Gamma(7/12)} \quad (31)$$

$$x^2 - 12 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/12)^2\Gamma(5/12)^2}{\Gamma(1/4)^4} \quad (32)$$

$$12x^2 - 1 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/4)^4}{\Gamma(1/12)^2\Gamma(5/12)^2} \quad (33)$$

3 Cubic Equations

$$x^3 - 4 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/6)\Gamma(2/3)}{\Gamma(1/3)\Gamma(1/2)} \quad (34)$$

$$4x^3 - 1 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/3)\Gamma(1/2)}{\Gamma(1/6)\Gamma(2/3)} \quad (35)$$

$$x^3 - x^2 - 2x + 1 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/7)\Gamma(6/7)}{\Gamma(2/7)\Gamma(5/7)} \quad (36)$$

$$x^3 - x^2 - 2x + 1 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(3/7)\Gamma(4/7)}{\Gamma(1/7)\Gamma(6/7)} \quad (37)$$

$$x^3 - x^2 - 2x + 1 = 0 \Rightarrow x = \frac{\Gamma(3/14)\Gamma(5/14)}{\Gamma(1/14)\Gamma(1/2)} \quad (38)$$

$$x^3 - 2x^2 - x + 1 = 0 \Rightarrow x = \frac{\Gamma(1/7)\Gamma(6/7)}{\Gamma(3/7)\Gamma(4/7)} \quad (39)$$

$$x^3 - 2x^2 - x + 1 = 0 \Rightarrow x = \frac{\Gamma(1/14)\Gamma(1/2)}{\Gamma(3/14)\Gamma(5/14)} \quad (40)$$

$$x^3 - 2x^2 - x + 1 = 0 \Rightarrow x = \frac{\Gamma(2/7)\Gamma(5/7)}{\Gamma(1/7)\Gamma(6/7)} \quad (41)$$

$$x^3 + x^2 - 2x - 1 = 0 \Rightarrow x = \frac{\Gamma(2/7)\Gamma(5/7)}{\Gamma(3/7)\Gamma(4/7)} \quad (42)$$

$$x^3 + 2x^2 - x - 1 = 0 \Rightarrow x = \frac{\Gamma(3/7)\Gamma(4/7)}{\Gamma(2/7)\Gamma(5/7)} \quad (43)$$

$$x^3 - 3x^2 - 2x + 2 = 0 \Rightarrow x = \frac{\Gamma(1/8)^2\Gamma(3/4)}{\Gamma(1/4)\Gamma(3/8)^2} \quad (44)$$

$$2x^3 - 2x^2 - 3x + 1 = 0 \Rightarrow x = \frac{\Gamma(1/4)\Gamma(3/8)^2}{\Gamma(1/8)^2\Gamma(3/4)} \quad (45)$$

$$x^3 - 3x - 1 = 0 \Rightarrow x = \frac{\Gamma(1/9)\Gamma(8/9)}{\Gamma(2/9)\Gamma(7/9)} \quad (46)$$

$$x^3 - 3x^2 + 3 = 0 \Rightarrow x = \frac{\Gamma(1/9)\Gamma(8/9)}{\Gamma(1/3)\Gamma(2/3)} \quad (47)$$

$$x^3 - 3x^2 + 3 = 0 \Rightarrow x = \frac{\Gamma(2/9)\Gamma(7/9)}{\Gamma(1/3)\Gamma(2/3)} \quad (48)$$

$$x^3 - 3x^2 + 1 = 0 \Rightarrow x = \frac{\Gamma(1/9)\Gamma(8/9)}{\Gamma(4/9)\Gamma(5/9)} \quad (49)$$

$$x^3 - 3x^2 + 1 = 0 \Rightarrow x = \frac{\Gamma(4/9)\Gamma(5/9)}{\Gamma(2/9)\Gamma(7/9)} \quad (50)$$

$$x^3 + 3x^2 - 1 = 0 \Rightarrow x = \frac{\Gamma(2/9)\Gamma(7/9)}{\Gamma(1/9)\Gamma(8/9)} \quad (51)$$

$$x^3 - 3x + 1 = 0 \Rightarrow x = \frac{\Gamma(2/9)\Gamma(7/9)}{\Gamma(4/9)\Gamma(5/9)} \quad (52)$$

$$x^3 - 3x + 1 = 0 \Rightarrow x = \frac{\Gamma(4/9)\Gamma(5/9)}{\Gamma(1/9)\Gamma(8/9)} \quad (53)$$

$$3x^3 - 3x + 1 = 0 \Rightarrow x = \frac{\Gamma(1/3)\Gamma(2/3)}{\Gamma(1/9)\Gamma(8/9)} \quad (54)$$

$$3x^3 - 3x + 1 = 0 \Rightarrow x = \frac{\Gamma(1/3)\Gamma(2/3)}{\Gamma(2/9)\Gamma(7/9)} \quad (55)$$

$$3x^3 - 3x - 1 = 0 \Rightarrow x = \frac{\Gamma(1/3)\Gamma(2/3)}{\Gamma(4/9)\Gamma(5/9)} \quad (56)$$

$$x^3 + 3x^2 - 3 = 0 \Rightarrow x = \frac{\Gamma(4/9)\Gamma(5/9)}{\Gamma(1/3)\Gamma(2/3)} \quad (57)$$

$$x^3 - 2x^2 - 2x + 1 = 0 \Rightarrow x = \frac{\Gamma(1/10)\Gamma(9/10)}{\Gamma(3/10)\Gamma(7/10)} \quad (58)$$

$$x^3 - 4x^2 + 3x + 1 = 0 \Rightarrow x = \frac{\Gamma(1/14)\Gamma(13/14)}{\Gamma(3/14)\Gamma(11/14)} \quad (59)$$

$$x^3 - 4x^2 - 4x + 8 = 0 \Rightarrow x = \frac{\Gamma(1/14)\Gamma(13/14)}{\Gamma(1/2)^2} \quad (60)$$

$$x^3 + 3x^2 - 4x + 1 = 0 \Rightarrow x = \frac{\Gamma(3/14)\Gamma(11/14)}{\Gamma(1/14)\Gamma(13/14)} \quad (61)$$

$$x^3 - 4x^2 + 3x + 1 = 0 \Rightarrow x = \frac{\Gamma(3/14)\Gamma(11/14)}{\Gamma(5/14)\Gamma(9/14)} \quad (62)$$

$$x^3 + 4x^2 - 4x - 8 = 0 \Rightarrow x = \frac{\Gamma(3/14)\Gamma(11/14)}{\Gamma(1/2)^2} \quad (63)$$

$$x^3 + 3x^2 - 4x + 1 = 0 \Rightarrow x = \frac{\Gamma(5/14)\Gamma(9/14)}{\Gamma(3/14)\Gamma(11/14)} \quad (64)$$

$$x^3 - 4x^2 - 4x + 8 = 0 \Rightarrow x = \frac{\Gamma(5/14)\Gamma(9/14)}{\Gamma(1/2)^2} \quad (65)$$

$$8x^3 - 4x^2 - 4x + 1 = 0 \Rightarrow x = \frac{\Gamma(1/2)^2}{\Gamma(1/14)\Gamma(13/14)} \quad (66)$$

$$8x^3 + 4x^2 - 4x - 1 = 0 \Rightarrow x = \frac{\Gamma(1/2)^2}{\Gamma(3/14)\Gamma(11/14)} \quad (67)$$

$$8x^3 - 4x^2 - 4x + 1 = 0 \Rightarrow x = \frac{\Gamma(1/2)^2}{\Gamma(5/14)\Gamma(9/14)} \quad (68)$$

$$x^3 - 2x^2 - 8x + 8 = 0 \Rightarrow x = \frac{\Gamma(1/14)\Gamma(11/14)}{\Gamma(5/14)\Gamma(1/2)} \quad (69)$$

$$x^3 + 2x^2 - 8x - 8 = 0 \Rightarrow x = \frac{\Gamma(1/14)\Gamma(9/14)}{\Gamma(3/14)\Gamma(1/2)} \quad (70)$$

$$8x^3 - 8x^2 - 2x + 1 = 0 \Rightarrow x = \frac{\Gamma(5/14)\Gamma(1/2)}{\Gamma(1/14)\Gamma(11/14)} \quad (71)$$

$$8x^3 + 8x^2 - 2x - 1 = 0 \Rightarrow x = \frac{\Gamma(3/14)\Gamma(1/2)}{\Gamma(1/14)\Gamma(9/14)} \quad (72)$$

$$x^3 - 8x^2 + 12x + 8 = 0 \Rightarrow x = \frac{\Gamma(1/14)^2\Gamma(9/14)}{\Gamma(3/14)^2\Gamma(5/14)} \quad (73)$$

$$8x^3 + 12x^2 - 8x + 1 = 0 \Rightarrow x = \frac{\Gamma(3/14)^2\Gamma(5/14)}{\Gamma(1/14)^2\Gamma(9/14)} \quad (74)$$

$$x^3 - 3x^2 - 4x - 1 = 0 \Rightarrow x = \frac{\Gamma(1/14)\Gamma(13/14)}{\Gamma(5/14)\Gamma(9/14)} \quad (75)$$

$$x^3 + 4x^2 + 3x - 1 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(5/14)\Gamma(9/14)}{\Gamma(1/14)\Gamma(13/14)} \quad (76)$$

$$x^3 - 9x + 9 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/18)\Gamma(1/6)\Gamma(5/18)\Gamma(1/2)}{\Gamma(1/9)\Gamma(2/9)\Gamma(1/3)^2} \quad (77)$$

4 Quartic Equations

$$x^4 - 4x^2 - 4 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/8)^2\Gamma(3/4)}{\Gamma(1/8)\Gamma(3/8)\Gamma(1/2)} \quad (78)$$

$$x^4 - 8 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/8)\Gamma(5/8)}{\Gamma(1/4)\Gamma(1/2)} \quad (79)$$

$$4x^4 + 4x^2 - 1 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(3/8)\Gamma(1/2)}{\Gamma(1/8)\Gamma(3/4)} \quad (80)$$

$$x^4 - 2x^2 - 1 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/8)\Gamma(1/2)}{\Gamma(1/4)\Gamma(3/8)} \quad (81)$$

$$x^4 + 2x^2 - 1 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/4)\Gamma(3/8)}{\Gamma(1/8)\Gamma(1/2)} \quad (82)$$

$$8x^4 - 1 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/4)\Gamma(1/2)}{\Gamma(1/8)\Gamma(5/8)} \quad (83)$$

$$x^4 - 10x^2 + 5 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/10)\Gamma(9/10)}{\Gamma(2/5)\Gamma(3/5)} \quad (84)$$

$$x^4 - 10x^2 + 5 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(3/10)\Gamma(7/10)}{\Gamma(1/5)\Gamma(4/5)} \quad (85)$$

$$5x^4 - 10x^2 + 1 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/5)\Gamma(4/5)}{\Gamma(3/10)\Gamma(7/10)} \quad (86)$$

$$5x^4 - 10x^2 + 1 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(2/5)\Gamma(3/5)}{\Gamma(1/10)\Gamma(9/10)} \quad (87)$$

$$x^4 - 4x^2 + 1 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/12)\Gamma(11/12)}{\Gamma(1/6)\Gamma(5/6)} \quad (88)$$

$$x^4 - 4x^2 + 1 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/6)\Gamma(5/6)}{\Gamma(1/12)\Gamma(11/12)} \quad (89)$$

$$x^4 - 4x^2 + 1 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/6)\Gamma(5/6)}{\Gamma(5/12)\Gamma(7/12)} \quad (90)$$

$$x^4 - 4x^2 + 1 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(5/12)\Gamma(7/12)}{\Gamma(1/6)\Gamma(5/6)} \quad (91)$$

$$x^4 - 6x^2 - 3 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/12)\Gamma(3/4)}{\Gamma(1/4)\Gamma(7/12)} \quad (92)$$

$$x^4 - 12 = 0 \quad \Rightarrow \quad x = \frac{\Gamma(1/12)\Gamma(5/12)}{\Gamma(1/4)^2} \quad (93)$$

$$3x^4 + 6x^2 - 1 = 0 \Rightarrow x = \frac{\Gamma(1/4)\Gamma(7/12)}{\Gamma(1/12)\Gamma(3/4)} \quad (94)$$

$$12x^4 - 1 = 0 \Rightarrow x = \frac{\Gamma(1/4)^2}{\Gamma(1/12)\Gamma(5/12)} \quad (95)$$

$$x^4 - 36x^2 - 108 = 0 \Rightarrow x = \frac{\Gamma(1/12)^2\Gamma(3/4)}{\Gamma(1/4)\Gamma(1/3)^2} \quad (96)$$

$$x^4 - 24x^2 - 48 = 0 \Rightarrow x = \frac{\Gamma(1/12)^2\Gamma(2/3)}{\Gamma(1/4)^2\Gamma(1/3)} \quad (97)$$

$$108x^4 + 36x^2 - 1 = 0 \Rightarrow x = \frac{\Gamma(1/4)\Gamma(1/3)^2}{\Gamma(1/12)^2\Gamma(3/4)} \quad (98)$$

$$48x^4 + 24x^2 - 1 = 0 \Rightarrow x = \frac{\Gamma(1/4)^2\Gamma(1/3)}{\Gamma(1/12)^2\Gamma(2/3)} \quad (99)$$

$$x^4 + x^3 - 4x^2 - 4x + 1 = 0 \Rightarrow x = \frac{\Gamma(1/15)\Gamma(14/15)}{\Gamma(2/15)\Gamma(13/15)} \quad (100)$$

$$x^4 + x^3 - 4x^2 - 4x + 1 = 0 \Rightarrow x = \frac{\Gamma(7/15)\Gamma(8/15)}{\Gamma(1/15)\Gamma(14/15)} \quad (101)$$

$$x^4 - x^3 - 9x^2 - x + 1 = 0 \Rightarrow x = \frac{\Gamma(1/15)\Gamma(14/15)}{\Gamma(4/15)\Gamma(11/15)} \quad (102)$$

$$x^4 - x^3 - 9x^2 - x + 1 = 0 \Rightarrow x = \frac{\Gamma(4/15)\Gamma(11/15)}{\Gamma(1/15)\Gamma(14/15)} \quad (103)$$

$$x^4 - 6x^3 + 6x^2 + 9x - 9 = 0 \Rightarrow x = \frac{\Gamma(1/15)\Gamma(14/15)}{\Gamma(1/3)\Gamma(2/3)} \quad (104)$$

$$x^4 - 4x^3 - 4x^2 + x + 1 = 0 \Rightarrow x = \frac{\Gamma(1/15)\Gamma(14/15)}{\Gamma(7/15)\Gamma(8/15)} \quad (105)$$

$$x^4 - 4x^3 - 4x^2 + x + 1 = 0 \Rightarrow x = \frac{\Gamma(2/15)\Gamma(13/15)}{\Gamma(1/15)\Gamma(14/15)} \quad (106)$$

$$x^4 - x^3 - 4x^2 + 4x + 1 = 0 \Rightarrow x = \frac{\Gamma(2/15)\Gamma(13/15)}{\Gamma(4/15)\Gamma(11/15)} \quad (107)$$

$$x^4 - 5x^3 + 5x^2 + 5x - 5 = 0 \Rightarrow x = \frac{\Gamma(2/15)\Gamma(13/15)}{\Gamma(2/5)\Gamma(3/5)} \quad (108)$$

$$x^4 + x^3 - 9x^2 + x + 1 = 0 \Rightarrow x = \frac{\Gamma(2/15)\Gamma(13/15)}{\Gamma(7/15)\Gamma(8/15)} \quad (109)$$

$$x^4 + x^3 - 9x^2 + x + 1 = 0 \Rightarrow x = \frac{\Gamma(7/15)\Gamma(8/15)}{\Gamma(2/15)\Gamma(13/15)} \quad (110)$$

$$5x^4 - 5x^3 - 5x^2 + 5x - 1 = 0 \Rightarrow x = \frac{\Gamma(1/5)\Gamma(4/5)}{\Gamma(4/15)\Gamma(11/15)} \quad (111)$$

$$x^4 + 4x^3 - 4x^2 - x + 1 = 0 \Rightarrow x = \frac{\Gamma(4/15)\Gamma(11/15)}{\Gamma(2/15)\Gamma(13/15)} \quad (112)$$

$$x^4 + 4x^3 - 4x^2 - x + 1 = 0 \Rightarrow x = \frac{\Gamma(7/15)\Gamma(8/15)}{\Gamma(4/15)\Gamma(11/15)} \quad (113)$$

$$x^4 + 6x^3 + 6x^2 - 9x - 9 = 0 \Rightarrow x = \frac{\Gamma(4/15)\Gamma(11/15)}{\Gamma(1/3)\Gamma(2/3)} \quad (114)$$

$$x^4 - x^3 - 4x^2 + 4x + 1 = 0 \Rightarrow x = \frac{\Gamma(4/15)\Gamma(11/15)}{\Gamma(7/15)\Gamma(8/15)} \quad (115)$$

$$9x^4 - 9x^3 - 6x^2 + 6x - 1 = 0 \Rightarrow x = \frac{\Gamma(1/3)\Gamma(2/3)}{\Gamma(2/15)\Gamma(13/15)} \quad (116)$$

$$9x^4 - 9x^3 - 6x^2 + 6x - 1 = 0 \Rightarrow x = \frac{\Gamma(1/3)\Gamma(2/3)}{\Gamma(7/15)\Gamma(8/15)} \quad (117)$$

$$9x^4 + 9x^3 - 6x^2 - 6x - 1 = 0 \Rightarrow x = \frac{\Gamma(1/3)\Gamma(2/3)}{\Gamma(4/15)\Gamma(11/15)} \quad (118)$$

$$5x^4 - 5x^3 - 5x^2 + 5x - 1 = 0 \Rightarrow x = \frac{\Gamma(2/5)\Gamma(3/5)}{\Gamma(2/15)\Gamma(13/15)} \quad (119)$$

$$x^4 - 4x^3 - 14x^2 - 4x + 1 = 0 \Rightarrow x = \frac{\Gamma(1/20)\Gamma(19/20)}{\Gamma(9/20)\Gamma(11/20)} \quad (120)$$

$$x^4 - 4x^3 - 14x^2 - 4x + 1 = 0 \Rightarrow x = \frac{\Gamma(9/20)\Gamma(11/20)}{\Gamma(1/20)\Gamma(19/20)} \quad (121)$$

$$x^4 - 10x^3 + 30x^2 - 30x + 5 = 0 \Rightarrow x = \frac{\Gamma(1/20)^3\Gamma(1/5)^2\Gamma(9/20)}{\Gamma(1/10)^2\Gamma(3/20)^3\Gamma(7/20)} \quad (122)$$

$$5x^4 - 30x^3 + 30x^2 - 10x + 1 = 0 \Rightarrow x = \frac{\Gamma(1/10)^2\Gamma(3/20)^3\Gamma(7/20)}{\Gamma(1/20)^3\Gamma(1/5)^2\Gamma(9/20)} \quad (123)$$

5 Quintic Equations

$$x^5 - 8 = 0 \Rightarrow x = \frac{\Gamma(1/5)\Gamma(7/10)}{\Gamma(2/5)\Gamma(1/2)} \quad (124)$$

$$x^5 - 16 = 0 \Rightarrow x = \frac{\Gamma(1/10)\Gamma(3/5)}{\Gamma(1/5)\Gamma(1/2)} \quad (125)$$

$$8x^5 - 1 = 0 \Rightarrow x = \frac{\Gamma(2/5)\Gamma(1/2)}{\Gamma(1/5)\Gamma(7/10)} \quad (126)$$

$$16x^5 - 1 = 0 \Rightarrow x = \frac{\Gamma(1/5)\Gamma(1/2)}{\Gamma(1/10)\Gamma(3/5)} \quad (127)$$

$$x^5 + 2x^4 - 5x^3 - 2x^2 + 4x - 1 = 0 \Rightarrow x = \frac{\Gamma(3/11)\Gamma(8/11)}{\Gamma(1/11)\Gamma(10/11)} \quad (128)$$

$$x^5 - 2 = 0 \Rightarrow x = \frac{\Gamma(1/20)\Gamma(3/10)\Gamma(11/20)}{\Gamma(1/10)\Gamma(3/20)\Gamma(13/20)} \quad (129)$$

$$2x^5 - 1 = 0 \Rightarrow x = \frac{\Gamma(1/10)\Gamma(13/20)}{\Gamma(1/20)\Gamma(11/20)} \quad (130)$$